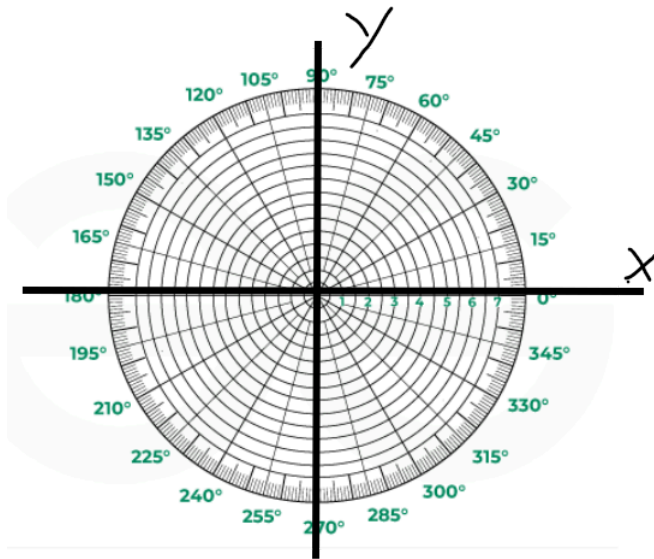


popquiz trig , vectors

 This is a preview of the published version of the quiz

Started: Apr 29 at 8:07pm

Quiz Instructions



Question 1 15 pts

1) The x component of vector $\vec{A}$ is 8.7 units, and its y component is -6.5 units. The magnitude of $\vec{A}$ is closest to

hint: in the coordinate system (X,Y) plot the point (8.7, -6.5). Trace an arrow from the origin to this point. You can use the polar graph (header)



11 units



9.9 units



12 units



8.9 units



7.9 units



Question 2 15 pts

building on previous question. Your vector should be in the 4th quadrant.

What is the angle between the vector and the positive x-axis/ Use tangent in the right angle triangle formed by the + xaxis and the vector. Use the magnitude angle = $\tan^{-1}$ (opp/adj)

hint: watch the sign. going CCW from +xaxis means positive angle. going CW from +axis means negative angle.

☐

27

☐

51

☐

- 51

☐

- 37

☐

Question 3 15 pts

) When rolled down a mountainside at 7.0 m/s, the horizontal component of its velocity vector was 1.8 m/s. What was the angle of the mountain surface above the horizontal? 2) _____

hint: I am helping you here (below). You are given the x-component and you see the hypotenuse is downslope so in the 4th quadrant.


☐

33

☐

75

☐

57

☐

15

⋮

Question 4 15 pts

6) A displacement vector is 34.0 m in length and is directed 60.0° east of north. Selecting from the choices in the table below, what are the components of this vector?

	Northward	Eastward
<u>choice</u>	<u>component</u>	<u>component</u>
1	29.4 m	17.0 m
2	18.2 m	28.1 m
3	22.4 m	11.5 m
4	17.0 m	29.4 m
5	25.2 m	18.2 m

hint: east of north is in the first quadrant. From +y-axis, sweep 60 degrees angle. you can use your polar graph

☐

5

☐

3

☐

2

☐

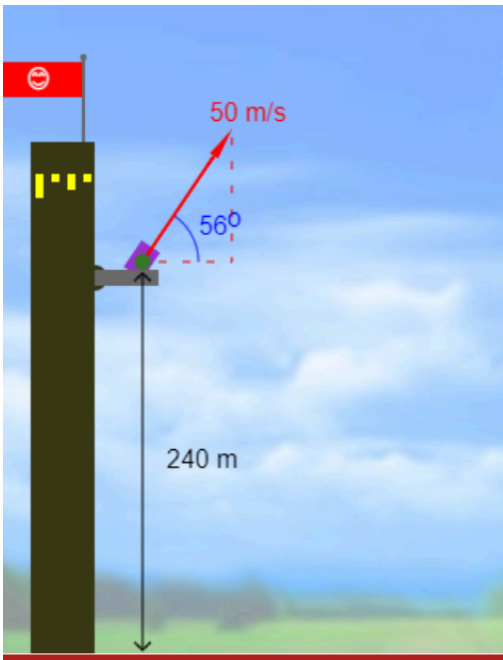
4

☐

1

⋮

Question 5 15 pts

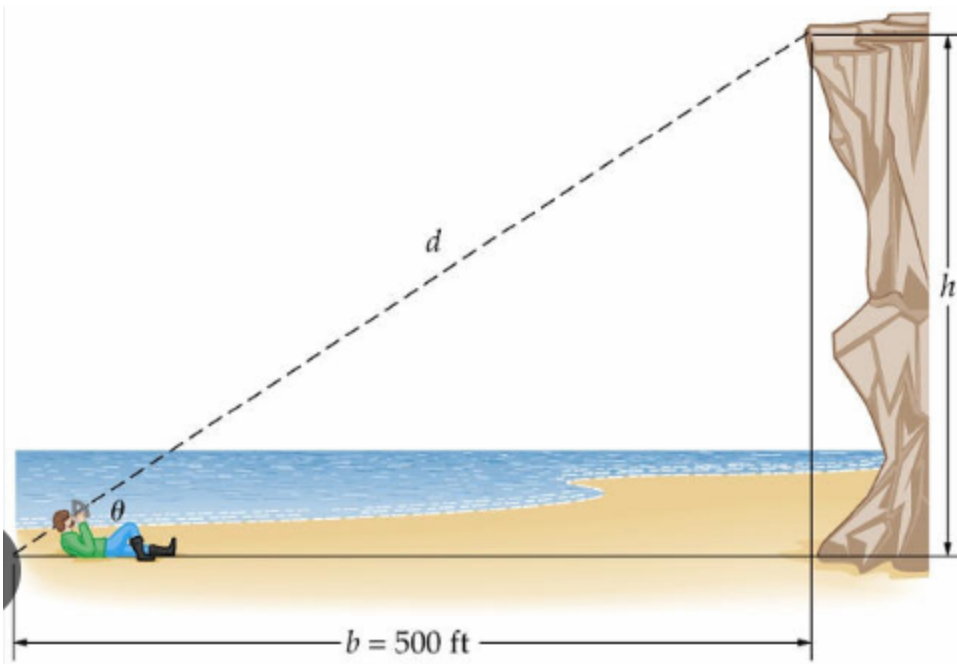


This projectile is ready to be launched. What is the y-component of the initial velocity V_{oy} ($t=0$)

- ☐ about 74m/s
- ☐ about 28m/s
- ☐ about 50m/s
- ☐ about 41m/s



Question 6 15 pts

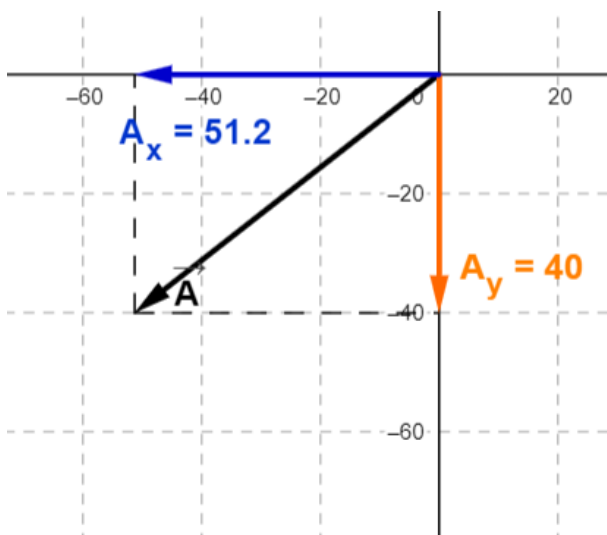


when you want to measure the height of a cliff, you can measure the angle from the horizontal to the top of the cliff and how far you are from the cliff. This is described in the Jules Verne Mysterious Island. This method is used in surveying. The angle here is 40 degrees. Round to the nearest 100

- ☐ about 300ft
- ☐ about 7200 ft
- ☐ about 500ft
- ☐ about 400 ft



Question 7 10 pts



What the coordinates of A in the (X,Y) system and its magnitude?

☐

A(51.2, 40) and 65

☐

A (-51.2, -40) and 32

☐

A(51.2, -40) and 4221

☐

A(-51.2, -40) and 65

Not saved

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